

AN INTRINSIC FACTOR OF AN ALTERNATING SUM INVOLVING CERTAIN RATIONAL FUNCTIONS

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Abstract

The main purpose of this paper is to give three types of proofs of the fact that there always exists a particular intrinsic factor in the numerator of an alternating sum with binomial coefficients involving certain rational functions.

1. Introduction

For an integer $n \geq 0$ and a non-zero variable x, one can easily find that

$$F(x) := \sum_{i=0}^{n} (-1)^i \binom{n}{i} \frac{1}{x^i} = \left(1 - \frac{1}{x}\right)^n = \frac{(x-1)^n}{x^n}. \tag{1.1}$$

So the numerator of $F(x)$ has the unique factor $(x-1)^n$, i.e., the x-intercept of $F(x)$ in a graph is the point $(1, 0)$ with multiplicity n. This almost self-evident example motivated us to become interested in alternating sums with binomial coefficients involving other rational functions that have the same property as the above.

Given any real number a different from -1, consider the slightly generalized alternating sum of $F(x)$ defined by

$$G_n(x; a) := \sum_{i=0}^{n} (-1)^i \binom{n}{i} \frac{1}{x^i + a}. \tag{1.2}$$

Thus $G_n(x; 0) = F(x)$. Multiplying this by $\prod_{i=0}^{n}(x^i + a)$, we get

$$H_n(x; a) := G_n(x; a) \prod_{i=0}^{n} (x^i + a) = \sum_{i=0}^{n} (-1)^i \binom{n}{i} \prod_{\substack{k=0 \\ k \neq i}}^{n} (x^k + a), \tag{1.3}$$

which is just the polynomial in x of degree $n(n+1)/2$. In particular, when $n = 0$, the empty product is defined to be 1 by convention.

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For the first few integers $n \geq 0$, we have by direct calculation,

$$\begin{aligned}
&H_0(x;a) = 1;\\
&H_1(x;a) = x-1;\\
&H_2(x;a) = (x-1)^2(x-a);\\
&H_3(x;a) = (x-1)^3\left(x^3-2ax^2-2ax+a^2\right);\\
&H_4(x;a) = (x-1)^4\left(x^6-3ax^5-5ax^4+3a(a-1)x^3+5a^2x^2+3a^2x-a^3\right).
\end{aligned}$$

These examples lead us to predicted that $(x-1)^n \mid H_n(x;a)$ may holds, i.e., the numerator of $G_n(x;a)$ has the factor $(x-1)^n$ as well as $F(x)$ in (1.1). As a result, this prediction was confirmed to be correct and so we can state the following.

Theorem 1.1. *The numerator of $G_n(x;a)$ has the factor $(x-1)^n$ for all $n \geq 0$.*

After trying various things, it turns out that proving this simple fact was not as easy as we initially imagined, at least for us, even though it seems like a high school level assignment at first glance. Given this situation, it is quite natural to think that we need to find any convincing rational proof of Theorem 1.1. This goal will be achieved in Section 2 by providing three different types of proofs from diverse points of views. In Section 3, we discuss an alternating sum with more terms than $G_n(x;a)$ in (1.2) that has exactly the same property as in Theorem 1.1. We conclude this paper, in Section 4, with additional discussion on an alternating sum that involves the reciprocals of polynomials with multiple real roots.

2. Three Different Types of Proofs of Theorem 1.1

The first proof is done by a straightforward induction on n, the second is a combinatorial proof using the elementary symmetric polynomials in algebra, and the third is a proof based on a certain geometric series expansion in calculus.

The First Proof. When $0 \leq n \leq 4$, the assertion holds from the examples given in the previous section. So assuming $(x-1)^n \mid H_n(x;a)$ for some $n \geq 5$, we will prove that $(x-1)^{n+1} \mid H_{n+1}(x;a)$ holds by induction on n. By replacing x^k+a in $H_n(x;a)$ with $x^{k+1}+a$ for each $k = 0, 1, \ldots, n$, consider the polynomial

$$g_n(x;a) := \sum_{i=0}^{n}(-1)^i\binom{n}{i}\prod_{\substack{k=0\\k\neq i}}^{n}(x^{k+1}+a).$$

Hereafter, let $h_n(x;a)$ denote the quotient of $H_n(x;a)$ divided by $(x-1)^n$, i.e., $H_n(x;a) = (x-1)^n h_n(x;a)$. Noting that each term in $g_n(x;a)$ is made up of a

product of n factors, we can express $g_n(x;a)$ for a non-zero variable x as follows:

$$\begin{aligned} g_n(x;a) &= x^n \sum_{i=0}^{n} (-1)^i \binom{n}{i} \prod_{\substack{k=0 \\ k\neq i}}^{n} \left(x^k + \frac{a}{x}\right) \\ &= x^n H_n\left(x; \frac{a}{x}\right) = x^n (x-1)^n h_n\left(x; \frac{a}{x}\right). \end{aligned} \tag{2.1}$$

On the other hand, using the identity $\binom{n+1}{i} = \binom{n}{i} + \binom{n}{i-1}$ with $\binom{n}{n+1} = \binom{n}{-1} = 0$ by convention, we get from (1.3),

$$\begin{aligned} H_{n+1}(x;a) &= \sum_{i=0}^{n+1} (-1)^i \binom{n+1}{i} \prod_{\substack{k=0 \\ k\neq i}}^{n+1} (x^k + a) \\ &= \sum_{i=0}^{n+1} (-1)^i \left(\binom{n}{i} + \binom{n}{i-1} \right) \prod_{\substack{k=0 \\ k\neq i}}^{n+1} (x^k + a) \\ &= \sum_{i=0}^{n} (-1)^i \binom{n}{i} \prod_{\substack{k=0 \\ k\neq i}}^{n+1} (x^k + a) + \sum_{i=1}^{n+1} (-1)^i \binom{n}{i-1} \prod_{\substack{k=0 \\ k\neq i}}^{n+1} (x^k + a). \end{aligned}$$

We now denote, simply by $C_1(x)$ and $C_2(x)$, the first and second sums on the last side, respectively. Since each term in $C_1(x)$ has the common factor $x^{n+1} + a$, we can write $C_1(x)$ under the above assumption as

$$\begin{aligned} C_1(x) &= (x^{n+1} + a) \sum_{i=0}^{n} (-1)^i \binom{n}{i} \prod_{\substack{k=0 \\ k\neq i}}^{n} (x^k + a) \\ &= (x^{n+1} + a) H_n(x;a) \\ &= (x^{n+1} + a)(x-1)^n h_n(x;a). \end{aligned}$$

Meanwhile, since each term in $C_2(x)$ has the common factor $1 + a$, by changing the product index notation, we are able to get from (2.1) and the assumption,

$$\begin{aligned} C_2(x) &= (1+a) \sum_{j=0}^{n} (-1)^{j+1} \binom{n}{j} \prod_{\substack{k=1 \\ k\neq j+1}}^{n+1} (x^k + a) \\ &= -(1+a) x^n \sum_{j=0}^{n} (-1)^j \binom{n}{j} \prod_{\substack{\ell=0 \\ \ell\neq j}}^{n} \left(x^\ell + \frac{a}{x}\right) \\ &= -(1+a) g_n(x;a) = -(1+a) x^n H_n\left(x; \frac{a}{x}\right) \\ &= -(1+a) x^n (x-1)^n h_n\left(x; \frac{a}{x}\right). \end{aligned}$$

Therefore, by adding $C_1(x)$ and $C_2(x)$, we finally obtain

$$\begin{aligned} H_{n+1}(x;a) &= C_1(x) + C_2(x) \\ &= (x-1)^n \left((x^{n+1}+a)h_n(x;a) - (1+a)x^n h_n\left(x;\frac{a}{x}\right)\right). \end{aligned}$$

Here we see that the part enclosed in the big parentheses on the last side has the factor $x-1$, because it disappears if we set $x=1$. Therefore, it was shown that $(x-1)^{n+1} \mid H_{n+1}(x;a)$ holds true and the proof is now complete. □

The above proof may seem a bit complicated and confusing, but this is the best we can do now as an inductive proof. The most notable feature of this proof is that it shows us how $H_{n+1}(x;a)$ is constructed from $H_n(x;a)$.

The Second Poof. Putting formally

$$t_i := x^i + a \quad \text{for} \quad i = 0, 1, \ldots, n,$$

consider the elementary symmetric polynomials in $n+1$ variables $\boldsymbol{t}:= (t_0, t_1, \ldots, t_n)$ written simply as $s_j(\boldsymbol{t})$, $j = 0, 1, \ldots, n$, defined by

$$\begin{aligned} s_0 &= s_0(\boldsymbol{t}) := 1; \\ s_j &= s_j(\boldsymbol{t}) := \sum_{0 \le i_1 < \cdots < i_j \le n} t_{i_1} \cdots t_{i_j}. \end{aligned}$$

That is, the polynomial s_j is the sum of all products of j distinct variables taken from the set $\{t_0, t_1, \ldots, t_n\}$. In particular, let $s_0 = 1$ by convention. Next, consider the elementary symmetric polynomials in n variables $\boldsymbol{t^{(i)}}:= (t_0, \ldots, t_{i-1}, t_{i+1}, \ldots, t_n)$ obtained from $\boldsymbol{t}$ by removing t_i, namely

$$\begin{aligned} s_0^{(i)} &= s_0(\boldsymbol{t^{(i)}}) := 1; \\ s_j^{(i)} &= s_j(\boldsymbol{t^{(i)}}) := \sum_{\substack{0 \le i_1 < \cdots < i_j \le n \\ i_k \ne i \text{ for } k=1,2,\ldots,n}} t_{i_1} \cdots t_{i_j} \quad (1 \le j \le n). \end{aligned} \tag{2.2}$$

By observing the removed variable t_i, we see that $s_j^{(i)}$ can be expressed in terms of s_k, $k = 0, 1, \ldots, j$, as follows:

$$s_j^{(i)} = \sum_{k=0}^{j} (-1)^k t_i^k s_{j-k} \quad \text{(see also Remark 1 given below).} \tag{2.3}$$

We now try to express $H_n(x;a)$ in (1.3) in terms of the elementary symmetric polynomials in (2.2). As is easily seen, it follows from (2.3) that

$$\begin{aligned} \sum_{i=0}^{n} (-1)^i \binom{n}{i} s_n^{(i)} &= \sum_{i=0}^{n} (-1)^i \binom{n}{i} \left(\sum_{k=0}^{n} (-1)^k t_i^k s_{n-k} \right) \\ &= \sum_{k=0}^{n} (-1)^k s_{n-k} \left(\sum_{i=0}^{n} (-1)^i \binom{n}{i} t_i^k \right). \end{aligned} \tag{2.4}$$

The inner sum in the last line of Equation (2.4) can be written, for any fixed k, as

$$\begin{aligned}\sum_{i=0}^{n}(-1)^i\binom{n}{i}t_i^k &= \sum_{i=0}^{n}(-1)^i\binom{n}{i}(x^i+a)^k\\ &= \sum_{i=0}^{n}(-1)^i\binom{n}{i}\sum_{r=0}^{k}\binom{k}{r}x^{ir}a^{k-r}\\ &= \sum_{r=0}^{k}\binom{k}{r}a^{k-r}\sum_{i=0}^{n}(-1)^i\binom{n}{i}x^{ir}\\ &= \sum_{r=0}^{k}\binom{k}{r}a^{k-r}(1-x^r)^n\\ &= (1-x)^n\sum_{r=0}^{k}\binom{k}{r}a^{k-r}\left(\frac{1-x^r}{1-x}\right)^n.\end{aligned}$$

Putting this back into (2.4), it turns out that $(x-1)^n$ is a factor of $H_n(x;a)$ and so the proof is complete. □

As we have seen in the above proof, Identity (2.3) plays a key role in achieving the goal. It is not necessary here, but the elementary symmetric polynomials formed by removing any number of variables from $\boldsymbol{t}$ can also be expressed in terms of s_j, $j=0,1,\ldots,n$. For the basic properties of these polynomials and many applications, see, e.g., the excellent books by Egge [1] and Macdonald [2].

The following remark provides a justification for Identity (2.3).

Remark 1. Identity (2.3) can be easily derived by the following steps;

(a) first, subtract the sum of all terms involving t_j from s_j;
(b) next, express the subtracted sum in (a) as the product of t_j and s_{j-1};
(c) repeat the same process as above until the result can be expressed as the product of t_j and s_0.

By following the above, the desired identity is finally obtained. For example, when $n=2$, one can express $s_2^{(1)}$ in terms of s_0, s_1, and s_2 as follows:

$$\begin{aligned}s_2^{(1)} &= t_0t_2 = (t_0t_1+t_0t_2+t_1t_2)-t_1(t_0+t_2)\\ &= s_2 - t_1((t_0+t_1+t_2)-t_1) = s_2 - t_1s_1 + t_1^2s_0,\end{aligned}$$

as indicated above.

The Third Proof. In what follows, we regard $G_n(x;a)$ in (1.2) as a rational function of two variables x and a with the domain consisting of all (x,a) pairs satisfying $x^i+a\neq 0$ for $i=0,1,\ldots,n$. As a matter of course, it is acceptable to discuss

$G_n(x;a)$ assuming that $a > 2$ and $|x-1| < 1/2$. Now, consider the geometric series expansion starting from $1/a$ and with the common ratio x^i/a such that

$$\frac{1}{x^i+a} = \frac{1}{a}\cdot\frac{1}{x^i/a+1} = \frac{1}{a}\sum_{j=0}^{\infty}(-1)^j\left(\frac{x^i}{a}\right)^j \quad (0 \le i \le n).$$

We apply this series in order to express $G_n(x;a)$ as

$$\begin{aligned} G_n(x;a) &= \sum_{i=0}^{n}(-1)^i\binom{n}{i}\sum_{j=0}^{\infty}\frac{(-1)^j}{a}\left(\frac{x^i}{a}\right)^j \\ &= \frac{1}{a}\sum_{j=0}^{\infty}\left(\frac{-1}{a}\right)^j\left(\sum_{i=0}^{n}(-1)^i\binom{n}{i}x^{ji}\right) \\ &= \frac{1}{a}\sum_{j=0}^{\infty}\left(\frac{-1}{a}\right)^j(1-x^j)^n \\ &= \frac{(1-x)^n}{a}\sum_{j=0}^{\infty}\left(\frac{-1}{a}\right)^j q_{j,n}(x), \end{aligned} \tag{2.5}$$

where $q_{j,n}(x)$ is the integer polynomial such that

$$q_{j,n}(x) := \left(\frac{1-x^j}{1-x}\right)^n \quad (j \ge 0).$$

Choosing a and x from the above ranges, we see that the series on the last line of Equation (2.5) is absolutely convergent and so the assertion holds true. □

The above proof is very interesting in that it only requires basic knowledge in calculus to solve a polynomial factor problem such as Theorem 1.1.

Each of the three proofs given above has its own characteristic, but there may be other simpler and more elegant proofs.

3. Alternating Sums with the Same Property as in Theorem 1.1

Let f be any given non-constant and real-rooted polynomial in $\mathbb{R}[x]$ that does not have a root of 1. Now consider the alternating sum defined by

$$Q_n(x;f) := \sum_{i=0}^{n}(-1)^i\binom{n}{i}\frac{1}{f(x^i)}. \tag{3.1}$$

In particular, if $f(x) = x + a$ ($a \neq -1$), then $Q_n(x;f)$ becomes $G_n(x;a)$ in (1.2).

Now assume that f has only the distinct single roots $\alpha_1, \alpha_2, \ldots, \alpha_m$ different from 1, where $m := \deg f$. That is, after dividing by the leading coefficient,

$$f(x) = \prod_{r=1}^{m} (x - \alpha_r), \quad \text{where} \quad \alpha_i \neq \alpha_j \ \text{ if } \ i \neq j. \tag{3.2}$$

By decomposing $1/f$ into partial fractions, we get

$$\frac{1}{f(x)} = \prod_{r=1}^{m} \frac{1}{x - \alpha_r} = \sum_{r=1}^{m} \frac{A_r}{x - \alpha_r},$$

where A_r, $r = 1, 2, \ldots, m$, are the uniquely determined constants. Therefore, using $G_n(x; a)$ in (1.2), the above $Q_n(x; f)$ can be expressed as

$$\begin{aligned} Q_n(x; f) &= \sum_{i=0}^{n} (-1)^i \binom{n}{i} \sum_{r=1}^{m} \frac{A_r}{x^i - \alpha_r} \\ &= \sum_{r=1}^{m} A_r \left(\sum_{i=0}^{n} (-1)^i \binom{n}{i} \frac{1}{x^i - \alpha_r} \right) \\ &= \sum_{r=1}^{m} A_r G_n(x; -\alpha_r). \end{aligned}$$

Since Theorem 1.1 tells us that the numerator of $G_n(x; -\alpha_r)$ has $(x-1)^n$ as a factor for each $r = 1, 2, \ldots, m$, we can state the following result.

Theorem 3.1. *The numerator of $Q_n(x; f)$ has the factor $(x - 1)^n$ for all $n \geq 0$.*

When $\deg f = 1$, the above result is consistent with Theorem 1.1. Therefore, Theorems 1.1 and 3.1 are effectively equivalent. At this point, we do not know if Theorem 3.1 applies only to the polynomials f of the form in (3.2). Concerning this question, see the conjecture stated in the next section.

4. Additional Discussion

Let f be a real-rooted monic polynomial of degree $d > 0$ that has no root of 1. Classifying the roots of f into simple roots α_r, $r = 1, 2, \ldots, m$, and multiple roots β_s, $s = 1, 2, \ldots, \ell$, let us write f as $f = uv$, where

$$u(x) := \prod_{r=1}^{m} (x - \alpha_r) \quad \text{and} \quad v(x) := \prod_{s=1}^{\ell} (x - \beta_s)^{q_s} \quad (q_s \geq 2),$$

so $d = m + \sum_{s=1}^{\ell} q_s$. Below, we will show that the numerator of the alternating sum involving $1/f$ has at least the factor $x - 1$.

By performing a partial fraction decomposition on $1/f$, we have

$$\begin{aligned}\frac{1}{f(x)} &= \left(\prod_{r=1}^{m}\frac{1}{x-\alpha_r}\right)\left(\prod_{s=1}^{\ell}\frac{1}{(x-\beta_s)^{q_s}}\right)\\ &= \sum_{r=1}^{m}\frac{A_r}{x-\alpha_r}+\sum_{s=1}^{\ell}\left(\sum_{j=1}^{q_s}\frac{B_{sq_j}}{(x-\beta_s)^j}\right)\\ &= \left\{\sum_{r=1}^{m}\frac{A_r}{x-\alpha_r}+\sum_{s=1}^{\ell}\frac{B_{sq_1}}{x-\beta_s}\right\}+\sum_{s=1}^{\ell}\sum_{j=2}^{q_s}\frac{B_{sq_j}}{(x-\beta_s)^j},\end{aligned}$$

where the A's and B's are mutually dependent and uniquely determined constants. Therefore, the alternating sum we should consider now is

$$\sum_{i=0}^{n}(-1)^i\binom{n}{i}\frac{1}{f(x^i)} = \sum_{i=0}^{n}(-1)^i\binom{n}{i}S(x^i)+\sum_{s=1}^{\ell}\sum_{j=2}^{q_s}B_{sq_j}T_j(x^i), \tag{4.1}$$

where

$$S(x) := \sum_{r=1}^{m}\frac{A_r}{x-\alpha_r}+\sum_{s=1}^{\ell}\frac{B_{sq_1}}{x-\beta_s} \quad \text{and} \quad T_j(x) := \sum_{i=0}^{n}(-1)^i\binom{n}{i}\frac{1}{(x-\beta_s)^j}.$$

We see from Theorem 3.1 that the numerator of the first sum on the right-hand side of (4.1) has the factor $(x-1)^n$. On the other hand, multiplying $T_j(x^i)$ by $\prod_{i=0}^{n}(x^i-\beta_s)^j$ provides

$$W_j(x^i) := T_j(x^i)\prod_{i=0}^{n}(x^i-\beta_s)^j = \sum_{i=0}^{n}(-1)^i\binom{n}{i}\prod_{\substack{k=0\\k\neq i}}^{n}(x^k-\beta_s)^j,$$

which is the polynomial of degree $n(n+1)/2$. Putting here $x=1$ leads to

$$W_j(1) = (1-\beta_s)^{nj}\sum_{i=0}^{n}(-1)^i\binom{n}{i} = (1-\beta_s)^{nj}(1-1)^n = 0,$$

which implies $(x-1)\mid W_j(x^i)$ for all $i=0,1,\ldots,n$. Therefore, taking account of the previous result for $S(x)$, it becomes clear that the numerator of the alternating sum in (4.1) has at least the factor $x-1$, as initially claimed.

The above result may not be particularly remarkable, but it is all we currently know about the numerator of an alternating sum that involves the reciprocals of a real-rooted f. Of course, the situation is quite different when f has complex roots. Unfortunately, we do not know of a good way to find any non-trivial factors other than $x-1$, for example, such as a factor that is some power of $x-1$. It is not for sure, but perhaps we need an entirely different way to search for them.

At the end of this paper, we would like to present the following conjecture that arose from various experimental evidence.

Conjecture. Let f be a non-zero real-rooted polynomial in $\mathbb{R}[x]$. Then, the numerator of $Q_n(x; f)$ in (3.1) has 'at least' the factor $(x-1)^n$ if and only if f is either a monomial or a polynomial that has only single roots different from 1.

Typical examples of this conjecture apply to Theorems 1.1 and 3.1, as already mentioned. It should be noted that for a special value of a, there is a possibility that $(x-1)^N \mid H_n(x; a)$ holds for N greater than n. This phenomenon occurs only when $h_n(x; a)$, the quotient of $H_n(x; a)$ divided by $(x-1)^n$, has at least the factor $x-1$. For example, if $a = 2 \pm \sqrt{3}$, then we have $(x-1)^4 \mid H_3(x; a)$ because of $(x-1) \mid h_3(x; a)$. On the other hand, if the above conjecture is true, then the numerator of the alternating sum in (4.1) never has a higher-order factor of $x-1$.

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References

[1] E. S. Egge, *An Introduction to Symmetric Functions and Their Combinatorics*, volume 91 of Student Mathematical Library, Amer. Math. Soc., Providence, RI, 2019.

[2] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, Oxford Classic Texts in the Physical Sciences, 2nd ed., The Clarendon Press, Oxford University Press, New York, 2015.